Fractal Dimension: Theory and Computation

Box-Counting Estimation via Hilbert Curve Analysis

ME

May 25, 2026

Contents

1 Introduction

This is a test.

2 Measure Theory

The material in this section is standard and follows the presentation in [Men26].

First to define measure theory (I didn't take Lebesgue so I will be using the above notes...)

A measure is a way to assign a "size" to subsets of a set X. Let $\mathcal{F}$ be a σ-algebra on X.

Definition 2.1. (Measure). Let $(X, \mathcal{F})$ be a measurable space. A measure on (X, $\mathcal{F}$) is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

1. $\mu(\emptyset) = 0$
2. **Countable additivity:** if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

The triple $(X, \mathcal{F}, \mu)$ is called a **measure space**.

Proposition 2.2 (Basic properties)**.** *Let $(X, \mathcal{F}, \mu)$ be a measure space.*

1. ***Monotonicity:***

$$A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$$

2. ***Countable subadditivity:***

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

3. ***Continuity from below:*** *if $A_1 \subseteq A_2 \subseteq \cdots$ then*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

4. ***Continuity from above:*** *if $A_1 \supseteq A_2 \supseteq \cdots$ and $\mu(A_1) < \infty$ then*

$$\mu\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

The canonical example is the **Lebesgue measure** λ^n on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, where $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ-algebra — the smallest σ-algebra containing all open sets. Lebesgue measure assigns to each box $[a_1, b_1] \times \cdots \times [a_n, b_n]$ its volume $(b_1 - a_1) \cdots (b_n - a_n)$, and extends uniquely to all Borel sets. It formalises the intuitive notions of length, area, and volume.

The inadequacy of Lebesgue measure for fractal geometry is immediate: a curve $\gamma \subset \mathbb{R}^2$ has $\lambda^2(\gamma) = 0$ (zero area) but $\lambda^1(\gamma)$ may be infinite (infinite length). Neither gives useful information about the "size" of γ. What is needed is a family of measures parametrised by a continuous exponent $s \geq 0$.

To construct such a family, we need the notion of an outer measure, which is slightly weaker than a measure — it is defined on *all* subsets of X, not just measurable ones, and only requires subadditivity rather than full additivity.

Definition 2.3 (Outer measure)**.** A function $\mu^* : 2^X \to [0, \infty]$ is an **outer measure** on X if:

1. $\mu^*(\emptyset) = 0$
2. **Monotonicity**: $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$
3. **Countable subadditivity**:

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

The relationship between outer measures and measures is given by the Hahn-Carathéodory extension theorem: given an outer measure μ^* on X, the collection

$$\Sigma = \{A \subseteq X : \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \setminus A) \text{ for all } B \subseteq X\}$$

is a σ-algebra, and $\mu^*|_\Sigma$ is a measure. Sets in Σ are called **μ^*-measurable**. In particular, all Borel sets are μ^*-measurable whenever μ^* is constructed from a Borel-compatible pre-measure — which will be the case for Hausdorff measure.

The standard way to construct an outer measure is via coverings. If $\mathcal{K} \subseteq 2^X$ covers X (with $\emptyset \in \mathcal{K}$) and $\tilde{\mu} : \mathcal{K} \to [0, \infty]$ assigns sizes to sets in $\mathcal{K}$ with $\tilde{\mu}(\emptyset) = 0$, then

$$\mu^*(A) = \inf\left\{\sum_{i=1}^{\infty} \tilde{\mu}(K_i) \;\middle|\; K_i \in \mathcal{K},\ A \subseteq \bigcup_{i=1}^{\infty} K_i\right\}$$

defines an outer measure on X. This is precisely the construction used to define Hausdorff measure — with $\mathcal{K}$ being all subsets of diameter at most δ, and $\tilde{\mu}(U) = |U|^s$.

3 Hausdorff Measure

3.1 Definition

Let $(X,\, d)$ denote a metric space, $S \subseteq X$ a Borel set.
For a set $U \subseteq X$, write $|U| := \operatorname{diam}(U) = sup_{x,y \in U} d(x,y)$

Definition 3.1. Let $s \geq 0$, and $\delta > 0$. Define

$$\mathcal{H}^s_\delta(S) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^s \;\middle|\; S \subseteq \bigcup_{i=1}^{\infty} U_i,\ |U_i| \leq \delta \right\}$$

This is the infimum over all countable covers of S by sets of diameter at most δ.

As $\delta \to 0$, we allow fewer and fewer covers, so $\mathcal{H}^s_\delta(S)$ is non-decreasing in $1/\delta$. The limit

$$\mathcal{H}^s(S) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(S) \in [0, \infty]$$

is the s-dimensional Hausdorff measure of S. This is possibly $+\infty$.
$\mathcal{H}^s$ is an outer measure on X, and its restriction to the Borel σ-algebra is a measure.

3.2 Phase Transition

Observe that $\mathcal{H}^s(S)$ exhibits a phase transition in s:

Theorem 3.2. *If $\mathcal{H}^s(S) < \infty$ for some s, then $\mathcal{H}^t(S) = 0$ for all $t > s$.*

Proof. Suppose $\mathcal{H}^s(S) < \infty$. For any cover $\{U_i\}$ with $|U_i| \leq \delta$. For $t > s$:

$$\Sigma |U_i|^t = \Sigma |U_i|^{t-s} \cdot |U_i|^s \leq \delta^{t-s} \Sigma |U_i|^s$$

Taking the infimum over all covers gives

$$\mathcal{H}^t_\delta(S) \leq \delta^{t-s} \cdot \mathcal{H}^s_\delta(S) \leq \delta^{t-s} \cdot \mathcal{H}^s(S)$$

where the second inequality holds since $\mathcal{H}^s_\delta(S)$ is non-decreasing in $1/\delta$.
Now let $\delta \to 0$, $t - s > 0$, and $\mathcal{H}^s(S) < \infty$ is a fixed finite constant. So the right hand side goes to 0, hence $\mathcal{H}^t(S) = 0$. $\square$

Corollary 3.3. *There exists a unique value $d \in [0, \infty]$ such that*

$$\mathcal{H}^s(S) = \begin{cases} \infty & s < d \\ 0 & s > d \end{cases}$$

At $s = d$, the measure $\mathcal{H}^d(S)$ may take any value in $[0, +\infty]$.

4 Hausdorff Dimension

Definition 4.1. (Hausdorff Dimension). The **Hausdorff Dimension** of S is the critical exponent

$$\dim_{\mathcal{H}}(S) = \inf\{s \geq 0 : \mathcal{H}^s(S) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(S) = +\infty\}$$

4.1 Basic Properties

Proposition 4.2 (Basic properties)**.** *Hausdorff dimension satisfies the following:*

1. ***Monotonicity****:* $A \subseteq B \implies \dim_{\mathcal{H}}(A) \leq \dim_{\mathcal{H}}(B)$
2. ***Countable stability****:* $\dim_{\mathcal{H}}(\cup_{i=1}^{\infty} A_i) = \sup_i \dim_{\mathcal{H}}(A_i)$
3. ***Countable sets****: if S is countable,* $\dim_{\mathcal{H}}(S) = 0$.
4. ***Smooth manifolds*** [1]*: if $M \subset \mathbb{R}^n$ is a smooth k-dimensional submanifold, then* $\dim_{\mathcal{H}}(M) = k$.

[1] A smooth k-dimensional manifold is just a set that locally looks like $\mathbb{R}^k$

References

[Men26] Angeliki Menegaki. *Measure Theory and Integration: Lecture Notes MATH50006*. 2026.